Project: GOLD

IT4609 –Mini Project

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this Project report **“Project GOLD”** is the bonafide work of **Kishore A (312423205107)** and **Joseph Sundar Marshal FV (312423205092)** who carried out the IT4609 - Mini Project work under my supervision. Certified further that to best of my knowledge and belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or an award was conferred on an earlier on this or any other certificate.

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ABSTRACT

Gold stands out as a classic investment because it holds its value when the economy gets rocky. Yet, predicting gold prices isn't easy. The market's unpredictable swings, non-linear patterns, and long-term shifts make the task tough. Standard statistical tools and even single deep learning models usually fall short. They just can't keep up across different time frames.

That's where Project:GOLD comes in. This mini project introduces a gold price forecasting system that leans on an ensemble of cutting-edge time-series models—specifically, Chronos-T5 and N-HiTS. By combining these models, the system pushes prediction accuracy and reliability to a higher level. It's built for flexibility, handling forecasts from daily and weekly up to monthly and even five-year horizons.

To power these forecasts, the system collects historical gold price data, runs it through a feature engineering pipeline, and manages everything in a nimble SQLite database. It doesn't just rely on one model's output. Instead, it blends predictions from both models using a weighted ensemble, which cuts down on variance and helps the system generalize better.

Users interact with the forecasts through a dynamic Streamlit dashboard. You can visualize trends, tweak the gold weight in grams, convert currencies, and export your results as CSV files—all from the same interface.

Testing shows the ensemble approach clearly beats individual models when it comes to Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The system runs automatically, keeps costs down, and fits the needs of both investors and educators looking for sharp, real-world analysis.

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LIST OF SYMBOLS AND ABBREVIATIONS

ABBREVIATIONS	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
CSV	Comma-Separated Values
DXA	Drawing Exchange Format Units
FX	Foreign Exchange
INR	Indian Rupee
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
N-HiTS	Neural Hierarchical Interpolation for Time Series
RMSE	Root Mean Square Error
SQL	Structured Query Language
SQLite	SQL Database Engine
T5	Text-to-Text Transfer Transformer
UI	User Interface
USD	United States Dollar

UX	User Experience

CHAPTER – 1

INTRODUCTION

Gold's been a cornerstone of value and investment for centuries. Investors have always looked to it as a shield against inflation, currency swings, and the kind of instability that rattles financial markets. But gold isn't simple. Its price moves in ways that reflect a tangled web of global demand and economic signals. The result? Gold prices swing—sometimes wildly—and the challenge of predicting those swings keeps investors and analysts on their toes.

With Artificial Intelligence and Machine Learning, time-series forecasting has changed. We're no longer stuck with just statistical models. Deep learning can now hunt down non-linear trends and long-range dependencies that traditional methods miss. This project takes advantage of those breakthroughs. The goal: predict gold prices with more accuracy and less hassle.

1.1 SYSTEM OVERVIEW

PROJECT GOLD is built to do one thing well—forecast gold prices using deep learning. The system pulls in historical gold prices, runs them through a feature engineering pipeline, and stores everything in a SQLite database for easy handling.

The core of the system? Two advanced models: Chronos-T5 and N-HiTS. Each model churns out its own predictions. Instead of picking just one, PROJECT GOLD combines their forecasts using a weighted ensemble. This approach boosts accuracy and smooths out the noise. Finally, the system delivers results through a Streamlit dashboard, so users get a clear, interactive view of past trends and future projections.

1.2 PROBLEM STATEMENT

Accurately predicting gold prices isn’t easy. The market is volatile. The time series is non-linear and non-stationary. Prices react to all kinds of macroeconomic and geopolitical factors.

Old-school models—like ARIMA or basic neural networks—just don’t cut it. They overfit, or they can’t see far enough ahead. What’s needed is a robust, automated system that delivers reliable forecasts across both short and long time spans.

1.3 MOTIVATION

People invest in gold everywhere—individuals, banks, portfolio managers. Still, most rely on manual analysis or simple prediction tools that fall short. This project aims to build a system that does better. The goal is to offer:

Reliable predictions over multiple time horizons

No need for manual number-crunching

Zero operational cost, thanks to open-source tools

A dashboard that's genuinely useful for planning investments

1.4 OBJECTIVES

This project sets out to:

Build a gold price forecasting system with deep learning

Use ensemble forecasting to sharpen prediction accuracy

Support short-term and long-term forecasting

Automate everything from data collection to predictions and storage

Visualize results in an interactive dashboard

1.5 SCOPE OF THE PROJECT

Here's what's in scope:

Analyzing historical gold price data

Applying advanced time-series forecasting models

Using ensemble prediction strategies

Visualizing and exporting forecast results

The system focuses on daily gold price forecasting—no real-time or intraday predictions.

1.6 ORGANIZATION OF THE REPORT

This report is laid out as follows:

Chapter 1 covers the introduction, objectives, and scope.

Chapter 2 reviews the literature and related work.

Chapter 3 explains system analysis, requirements, and algorithm design.

References list all sources used.

CHAPTER 2

LITERATURE SURVEY

A comprehensive review of the literature reveals the evolution of research and techniques in the domain of financial time-series forecasting, particularly with respect to gold price prediction. Gold, as a commodity, has long intrigued researchers and practitioners due to its economic importance and notorious volatility. As a result, methods ranging from classic statistical modeling to the latest deep learning architectures have been explored and refined in pursuit of more accurate and reliable forecasts. This chapter systematically examines the principal methodologies employed for time-series and financial price forecasting, highlighting their theoretical underpinnings, practical strengths, and inherent limitations.

2.1 TRADITIONAL STATISTICAL FORECASTING MODELS

Historically, statistical models such as Moving Averages, Exponential Smoothing, and the Autoregressive Integrated Moving Average (ARIMA) family have provided the foundational toolkit for financial forecasting. These models operate under the assumption that the underlying data exhibits linearity and stationarity—conditions rarely met in real-world financial series, especially in the case of gold prices. Hyndman and Athanasopoulos underscore that while ARIMA and its variants are adept at capturing short-term autocorrelation and trend components, their effectiveness is constrained by their inability to accommodate the abrupt regime shifts, non-linear dependencies, and heteroskedasticity inherent in gold price movements. Moreover, these models often require frequent recalibration, domain expertise for parameter selection, and can quickly become obsolete as market dynamics evolve. Their reliance on past values also means that sudden, exogenous shocks—common in the gold market—are not effectively incorporated, resulting in diminished forecasting accuracy, particularly over longer horizons.

2.2 ARTIFICIAL NEURAL NETWORKS FOR FORECASTING

To address the limitations of linear models, Artificial Neural Networks (ANNs) emerged as a promising alternative due to their ability to model complex, non-linear relationships within data. Research by Zhang and collaborators demonstrated that ANNs could uncover hidden patterns and interactions missed by conventional statistical approaches, leading to improved performance in diverse financial forecasting contexts. These models, inspired by biological neural networks, can approximate any continuous function and adaptively learn from data. However, traditional feedforward ANNs process each input independently and lack a mechanism for temporal memory, which is critical for time-series tasks. As a result, while they can detect and model intricate non-linearities in short-term data, they struggle to account for sequential dependencies and long-range trends, thereby limiting their practical utility in forecasting assets like gold, where both recent and historical context matter.

2.3 LONG SHORT-TERM MEMORY (LSTM) NETWORKS

Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber, represent a significant advancement in time-series modeling, especially for sequential data with complex temporal dependencies. By incorporating memory cells and gating mechanisms, LSTMs can selectively retain or forget information over extended periods, addressing the vanishing gradient problem that hampers traditional recurrent networks. Fischer and Krauss provided empirical evidence that LSTMs consistently outperform classical models and basic neural networks in financial forecasting tasks, capturing both short-term fluctuations and longer-term cycles in asset prices. Nonetheless, LSTMs are not without challenges: they demand large volumes of high-quality data for effective training, involve substantial computational resources, and require careful hyperparameter tuning to avoid overfitting or underfitting. Their performance can also degrade when tasked with very long-range forecasts, as even advanced memory mechanisms may not fully capture the intricacies of distant market dynamics.

2.4 DEEP LEARNING MODELS FOR TIME-SERIES FORECASTING

The proliferation of deep learning has led to the exploration of more sophisticated architectures for time-series forecasting. Convolutional Neural Networks (CNNs), traditionally employed in image processing, have been adapted to capture local dependencies and multi-scale patterns in sequential data. Borovykh and colleagues demonstrated that CNNs, through their hierarchical feature extraction, can outperform both statistical and basic neural models in forecasting financial time series. Building on this, Challu et al. introduced N-HiTS, a neural hierarchical interpolation model designed specifically for multi-horizon forecasting. N-HiTS excels at simultaneously capturing short-term volatility and medium-term cycles, leveraging a modular architecture that facilitates both accuracy and interpretability. These advancements have broadened the scope of deep learning in finance, enabling models to generalize across different forecasting horizons and adapt to varying market conditions with greater flexibility.

2.5 FOUNDATION MODELS FOR TIME-SERIES DATA

A transformative development in the field is the advent of foundation models, which leverage massive datasets and transfer learning to achieve unprecedented generalization and adaptability. Ansari and collaborators proposed Chronos, based on the transformer architecture, which is renowned for its prowess in natural language processing and, increasingly, in sequential data analysis. Chronos-T5, a variant tailored for time-series forecasting, employs attention mechanisms to capture long-range dependencies and context across diverse domains. Its zero-shot learning capability allows it to generate accurate forecasts for previously unseen datasets without extensive retraining or domain-specific feature engineering. This adaptability, coupled with robust performance on long-horizon forecasts, positions foundation models as a paradigm shift in financial prediction—enabling practitioners to handle the ever-increasing complexity and dynamism of markets like gold with greater confidence and efficiency.

2.6 ENSEMBLE FORECASTING TECHNIQUES

Ensemble methods have emerged as a practical and effective strategy to enhance forecasting accuracy by integrating the strengths of multiple models. By combining predictions from diverse algorithms—statistical, neural, and deep learning—ensembles mitigate the idiosyncratic weaknesses of individual models and provide a more balanced, robust forecast. Makridakis and his team, through extensive forecasting competitions, have consistently shown that ensemble methods surpass the performance of any single model, particularly in volatile and uncertain environments. Techniques such as bagging, boosting, and stacking allow ensembles to adapt to changing data distributions, reduce the variance of predictions, and handle non-stationarity more effectively. This robustness is especially valuable in gold price forecasting, where market conditions are influenced by a complex interplay of macroeconomic factors, geopolitical events, and investor sentiment.

2.7 CONCLUSION

The literature unequivocally demonstrates that while traditional statistical models and isolated deep learning networks offer valuable insights, they are often inadequate for the multifaceted challenges posed by financial time-series data. Modern deep learning techniques, particularly those leveraging hierarchical architectures and large foundation models, have set new benchmarks for accuracy and adaptability. Ensemble approaches further amplify these benefits, providing resilience against model-specific biases and the unpredictability of financial markets.

Informed by these developments, the proposed forecasting system adopts an ensemble framework that synergistically combines Chronos-T5 and N-HiTS. By leveraging the transfer learning capabilities and long-range forecasting strength of Chronos-T5 alongside the multi-horizon, modular design of N-HiTS, the system aspires to deliver forecasts that are not only more precise, but also inherently robust and scalable. This integrated approach addresses the complexities of gold price prediction head-on, paving the way for more reliable decision-making in an ever-volatile market landscape.

CHAPTER 3

SYSTEM ANALYSIS

System analysis is the process of thoroughly examining the current approach, identifying its flaws, and envisioning a more efficient and effective solution. In this chapter, I will provide a comprehensive evaluation of existing gold price forecasting systems, highlight their shortcomings in today's dynamic market, and introduce the design principles and operational advantages of the newly proposed system. This discussion will also address the necessary requirements, the rationale behind technology selections, and offer an in-depth walkthrough of the core algorithmic workflow that powers the innovation.

3.1 EXISTING SYSTEM

Presently, the landscape of gold price prediction is predominantly dominated by traditional statistical methods and elementary machine learning models. These legacy approaches are fundamentally rooted in analyzing historical price patterns and often rest on the critical assumption that market data demonstrates linear and stationary behavior. Techniques such as moving averages, linear and polynomial regression, and ARIMA models are widely used, primarily because of their simplicity and interpretability.

However, these models come with significant limitations. Data gathering is often a manual and repetitive process, requiring users to periodically download, clean, and format historical price records. Model retraining is not automated, which means analysts have to regularly update the models to keep up with the latest market developments—a task that not only consumes valuable time and resources but also introduces the risk of human error or oversight.

Moreover, these methods generally offer only short-term forecasts and perform poorly in volatile or rapidly changing markets. They lack robustness against unexpected events such as geopolitical crises or sudden shifts in investor sentiment, which can cause abrupt price movements. The absence of built-in automation or scalability makes these systems unsuitable for handling increasing data volumes or adapting to evolving user needs. Additionally, most lack meaningful tools for data visualization and decision support, making it difficult for investors to glean actionable insights from raw predictions.

3.1.1 LIMITATIONS OF EXISTING SYSTEM

The limitations of the current generation of forecasting systems manifest in several critical ways:

* Accuracy drops significantly during periods of high volatility or market turbulence, undermining investor confidence.

* Incapable of detecting intricate, non-linear, or long-range dependencies in the data, thus failing to capture emerging or subtle market shifts.

* Forecast horizons are typically limited, offering little value for medium- or long-term planning.

* Require frequent manual intervention for updates and retraining, making the process labor-intensive.

* Maintenance is costly in terms of both time and computational resources, with little potential for automation.

* Visualization tools and decision-making aids are either rudimentary or non-existent, restricting the practical utility for end-users.

3.2 PROPOSED SYSTEM

To overcome these hurdles, PROJECT GOLD has been conceived—a cutting-edge solution that harnesses the power of deep learning ensemble models for gold price prediction. This system fundamentally rethinks the forecasting workflow by integrating advanced time-series modeling techniques that surpass the capabilities of legacy approaches.

PROJECT GOLD employs a hybrid ensemble of Chronos-T5 and N-HiTS—two state-of-the-art models designed specifically for time-series analysis. Chronos-T5 excels at capturing long-term patterns and dependencies, while N-HiTS is adept at modeling short- and medium-term fluctuations with high precision. By combining their outputs in a weighted fashion, the system is able to produce forecasts that are both accurate and robust across various market conditions.

Automation is at the heart of PROJECT GOLD. The system autonomously retrieves historical gold prices from reliable online sources, eliminating the need for manual data collection. Through an automated feature engineering pipeline, raw data is transformed into informative features that enhance model performance. All processed data is stored in an efficient, lightweight SQLite database, ensuring that the system remains scalable and easy to manage.

Once the data is prepared, the ensemble models generate their respective forecasts. These individual outputs are then intelligently integrated using a weighted ensemble method that minimizes prediction errors by dynamically adjusting the influence of each model based on recent performance. The ensemble's final forecasts are archived in the database and seamlessly presented to users via an interactive Streamlit dashboard. This dashboard not only visualizes predictions and historical trends, but also empowers users to customize forecasting parameters and export results for further analysis.

3.2.1 ADVANTAGES OF THE PROPOSED SYSTEM

PROJECT GOLD offers a host of tangible benefits over traditional methods:

* Significantly improved forecast accuracy, especially during volatile and unpredictable market periods, due to the synergistic power of ensemble deep learning.

* Ability to generate reliable predictions across both short-term and long-term horizons, meeting the diverse needs of different investor profiles.

* Automation permeates the entire workflow, from data acquisition to model retraining and prediction generation, freeing users from time-consuming manual tasks.

* The platform is built entirely on open-source technologies, resulting in zero licensing or operational costs and fostering community-driven enhancements.

* The use of SQLite ensures that the underlying database remains lightweight, fast, and capable of scaling with growing datasets without compromising performance.

* An intuitive and visually engaging dashboard, designed with investors and analysts in mind, makes it easy to interpret results, visualize trends, and take informed actions.

3.3 REQUIREMENT SPECIFICATION

To ensure seamless deployment and optimal performance, PROJECT GOLD requires the following environment:

3.3.1 SOFTWARE REQUIREMENTS

* Operating System: Compatible with both Windows and Linux platforms, offering flexibility for diverse user bases.

* Programming Language: Developed in Python 3.10 or later, leveraging its rich ecosystem and support for scientific computing.

* Database: Utilizes SQLite for efficient, lightweight, and self-contained data storage.

* Core libraries and frameworks:

* NumPy and Pandas for rapid numerical computation and data manipulation

* PyTorch for building and training deep learning models

* Streamlit for constructing interactive dashboards

* Plotly for creating dynamic, high-quality data visualizations

3.3.2 HARDWARE REQUIREMENTS

* Processor: Intel i5 or higher to ensure responsive model training and inference

* RAM: Minimum of 8 GB to handle model computations and data processing tasks

* Storage: At least 10 GB of free disk space to accommodate historical data and model artifacts

* Internet Connection: Necessary for automated retrieval of up-to-date gold price data and potential model updates

3.4 LANGUAGE SPECIFICATION

PROJECT GOLD is engineered using the following technologies:

* Python serves as the primary language, orchestrating every stage from data ingestion and preprocessing to model training and automation workflows. Its versatility and extensive library support make it ideal for complex machine learning tasks.

* SQL, via SQLite, underpins the storage and retrieval of historical data, engineered features, and prediction outputs, providing a robust yet lightweight database solution.

* Streamlit is employed to create a user-friendly, interactive dashboard that democratizes access to sophisticated forecasting tools, allowing even non-technical users to benefit from advanced analytics.

3.5 ALGORITHM DESCRIPTION

The operational backbone of PROJECT GOLD is a meticulously designed sequence of automated steps intended to maximize forecasting accuracy and user convenience:

1. The system autonomously fetches up-to-date historical gold prices from trusted financial data providers, ensuring a continuous and reliable data pipeline.

2. Retrieved data is systematically stored in an SQLite database, where it is organized and made accessible for downstream processing.

3. An automated feature engineering module transforms raw price data into a set of informative features, such as technical indicators and statistical summaries, that enhance model learning.

4. Chronos-T5 is deployed to analyze long-term patterns and produce extended horizon forecasts, drawing on its advanced sequence modeling capabilities.

5. In parallel, N-HiTS processes the same dataset, focusing on short- and medium-term trends to generate complementary predictions.

6. The outputs from both models are combined using a dynamically weighted ensemble technique. This method evaluates recent model performance, assigning greater influence to the model with superior accuracy, thereby optimizing the reliability of the final prediction.

7. The ensemble's output is stored back in the database for record-keeping and future analysis.

8. The Streamlit dashboard synthesizes all available information, presenting forecasts, historical trends, and interactive charts that allow users to explore various scenarios.

9. For added flexibility, users can export forecast results as CSV files, facilitating further analysis and integration into investment decision workflows.

Through this comprehensive, automated, and user-centric approach, PROJECT GOLD not only addresses the deficiencies of existing systems but also sets a new standard for gold price forecasting technology, making advanced predictive analytics accessible and actionable for all stakeholders.

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